



Topology Density Map for Urban Data Visualization and Analysis

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Agenda

Introduction

Motivation

Topology density map

- Requirement Analysis
- Accessibility
- Topology Density Map Construction

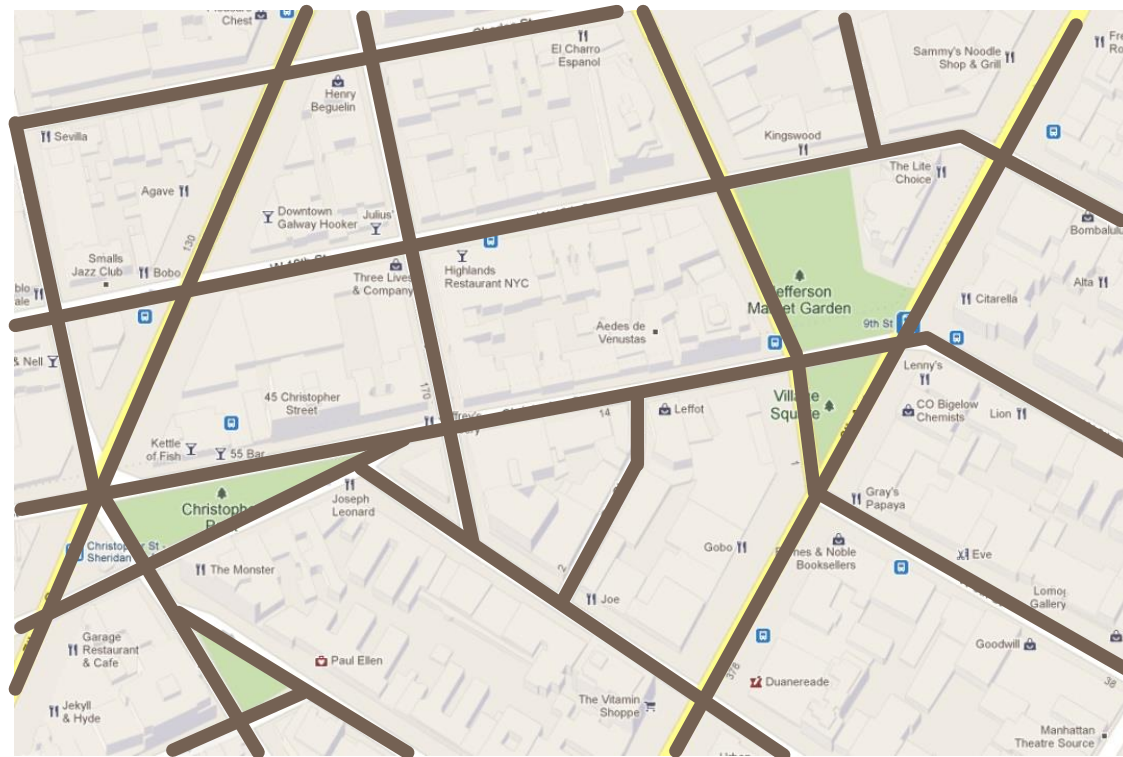
Evaluation

- Case Studies
- Computation Efficiency

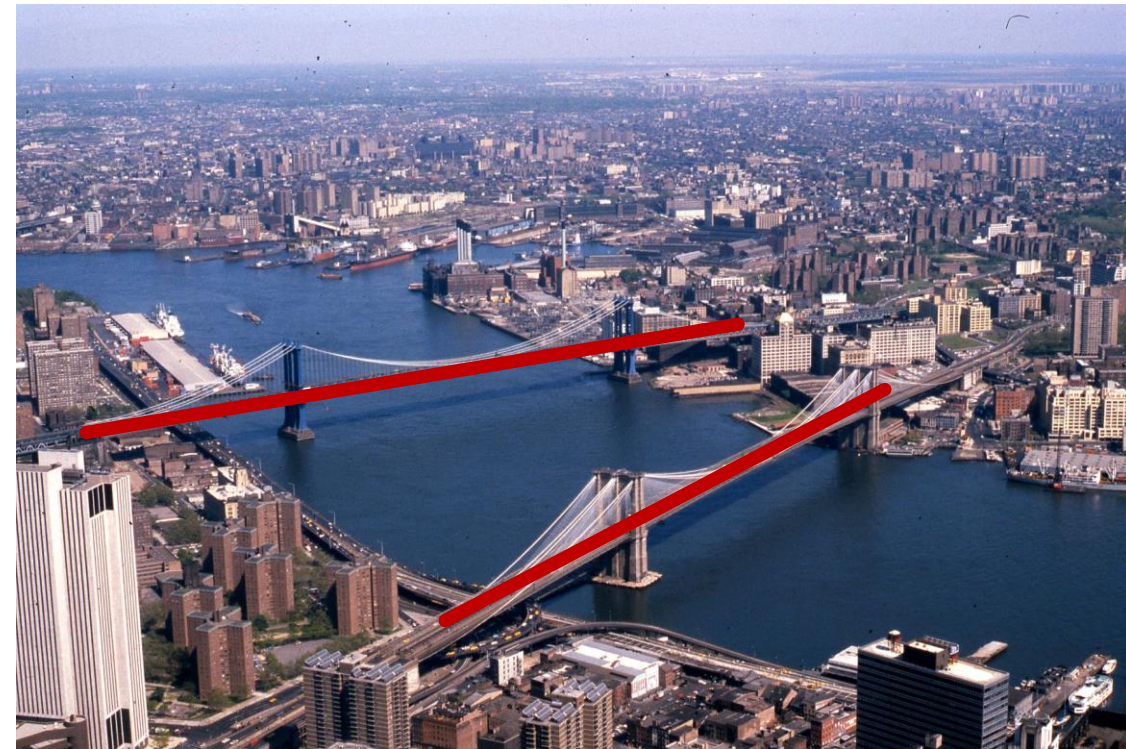
Conclusion

- Limitations & Future Work
- Contributions
- Take Home Messages

Road Structure

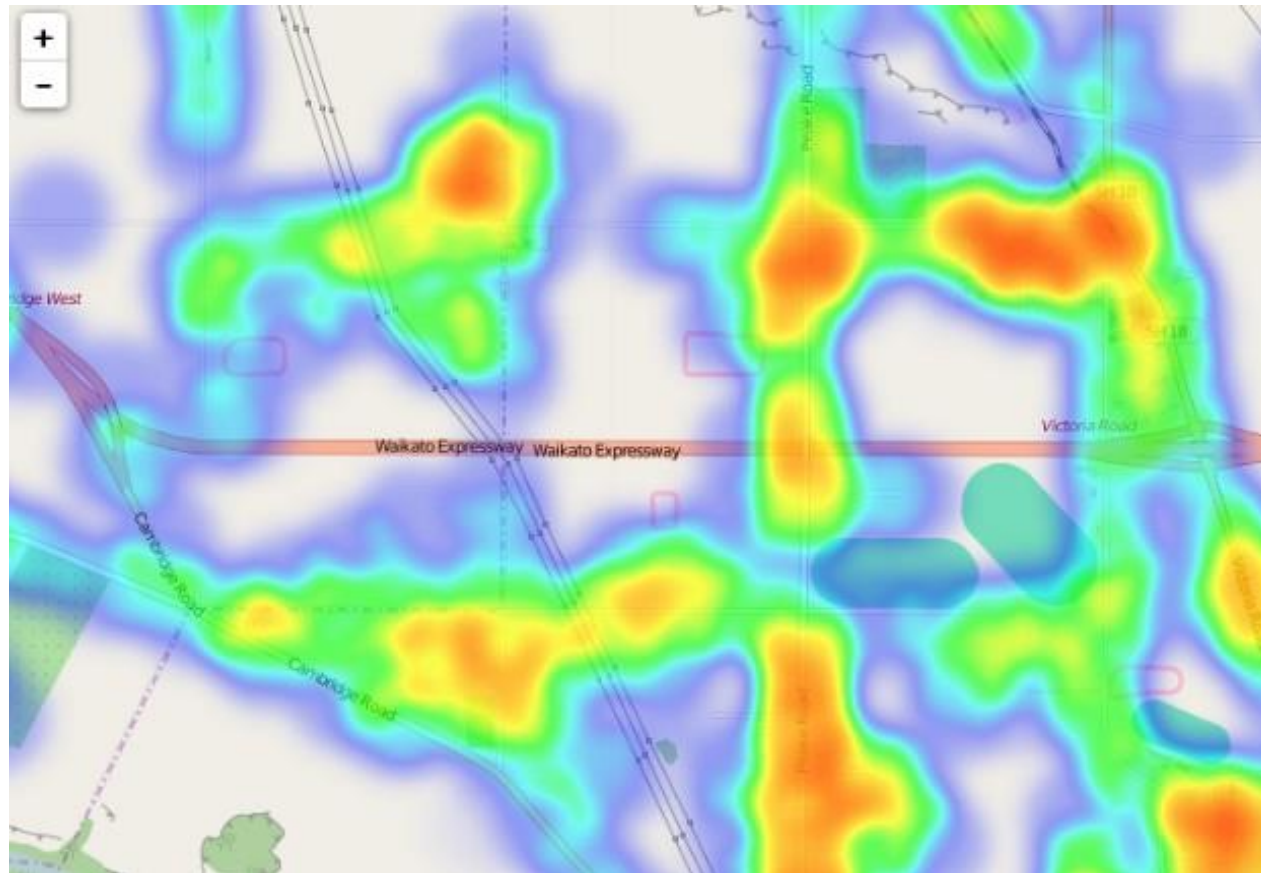


Road Topology



Road Connectivity

Density Map



- Depicts the continuous distribution of scalar field in a 2D planar space
- Widely used in urban analysis and urban planning
- The **K**ernel **D**ensity **E**stimation (KDE) is a common nonparametric model for density estimation

Motivation

➤ Topological:

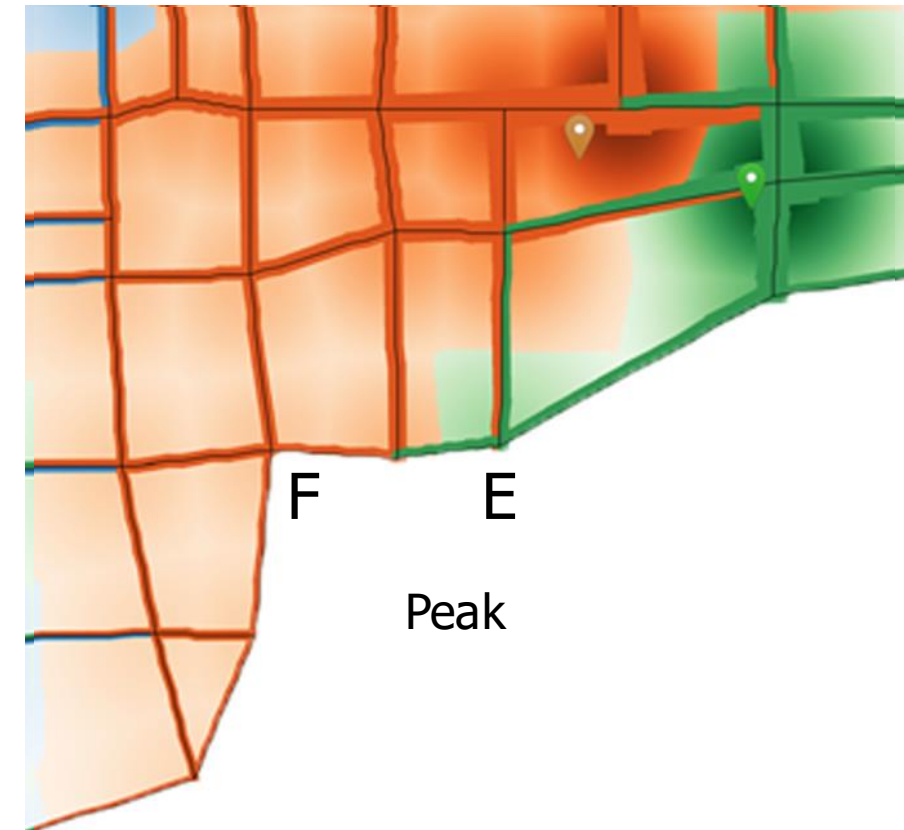
Locations in urban area are connected through roads and intersections

➤ Directional:

In most cities, a road has two different directions, unlike the Euclidean distance, the proximity of $A \rightarrow B$ could be very different from that of $B \rightarrow A$.

➤ Temporal Variant:

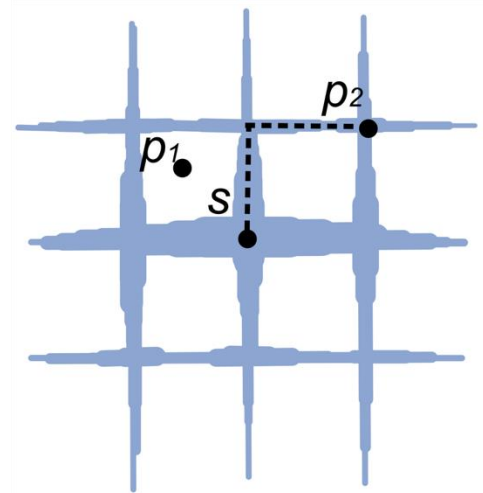
The access time between locations depends much on traffic conditions that vary over time



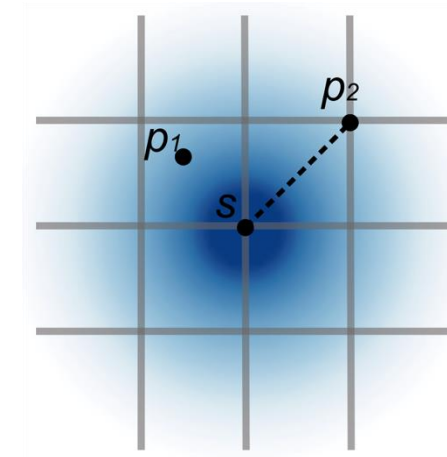
Requirement Analysis

R1: Correctness

- ✓ Traffic condition is dynamic and time variant
- ✓ Road networks in urban area exhibit a complex topology
- ✓ roads in a city are often bidirectional



NKDE



KDE

R2: Intuitiveness

- ✓ Reveal the density distribution on the 2D space

Accessibility

- POI accessibility is represented as scalar fields distributed over space
- The access time between a source POI and a target location:

$$Acc(P, H) = (\sum_{i=1}^n t(seg_i) + t(w))^{-\alpha}$$

Acc : POI accessibility

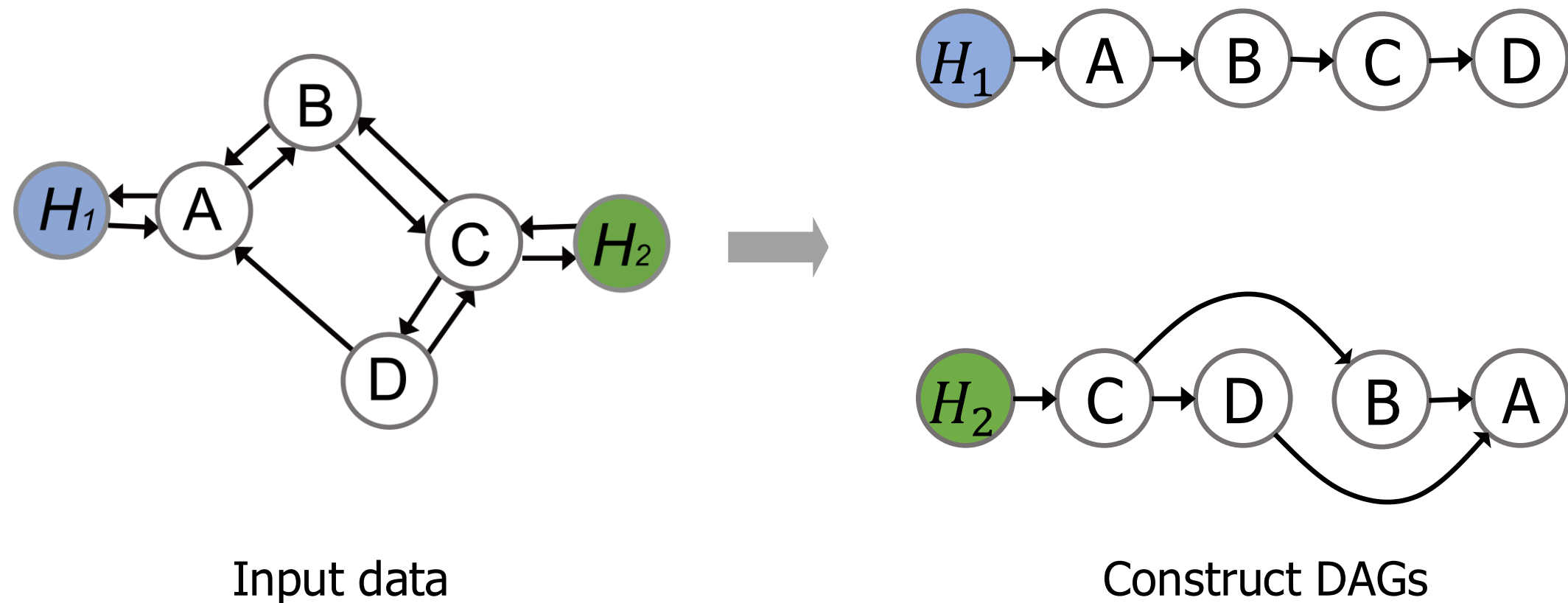
$t(\cdot)$: Vehicle access time

seg_i : Road segment

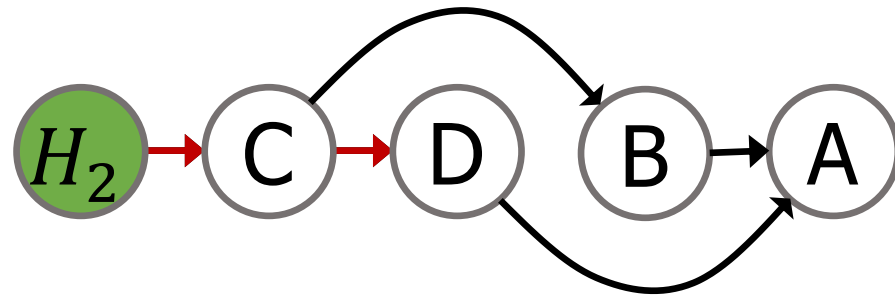
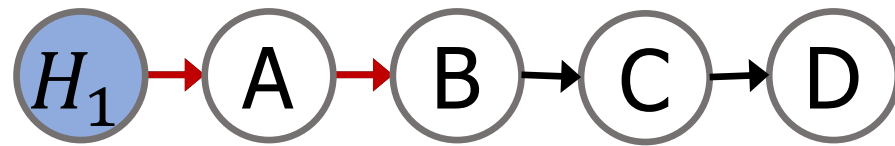
w : Walkway

α : Accessibility attenuation coefficient

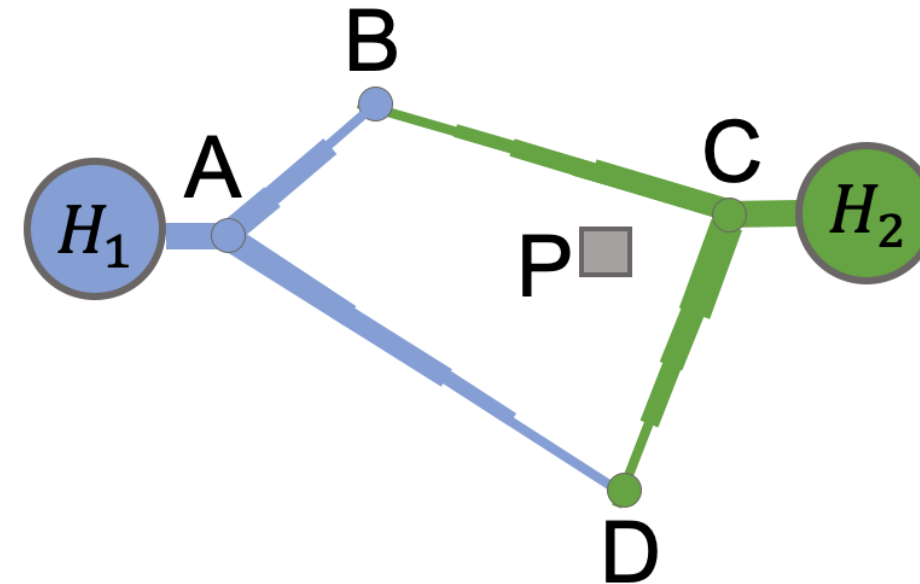
Density Estimation on Road Network



Density Estimation on Road Network Cont'd

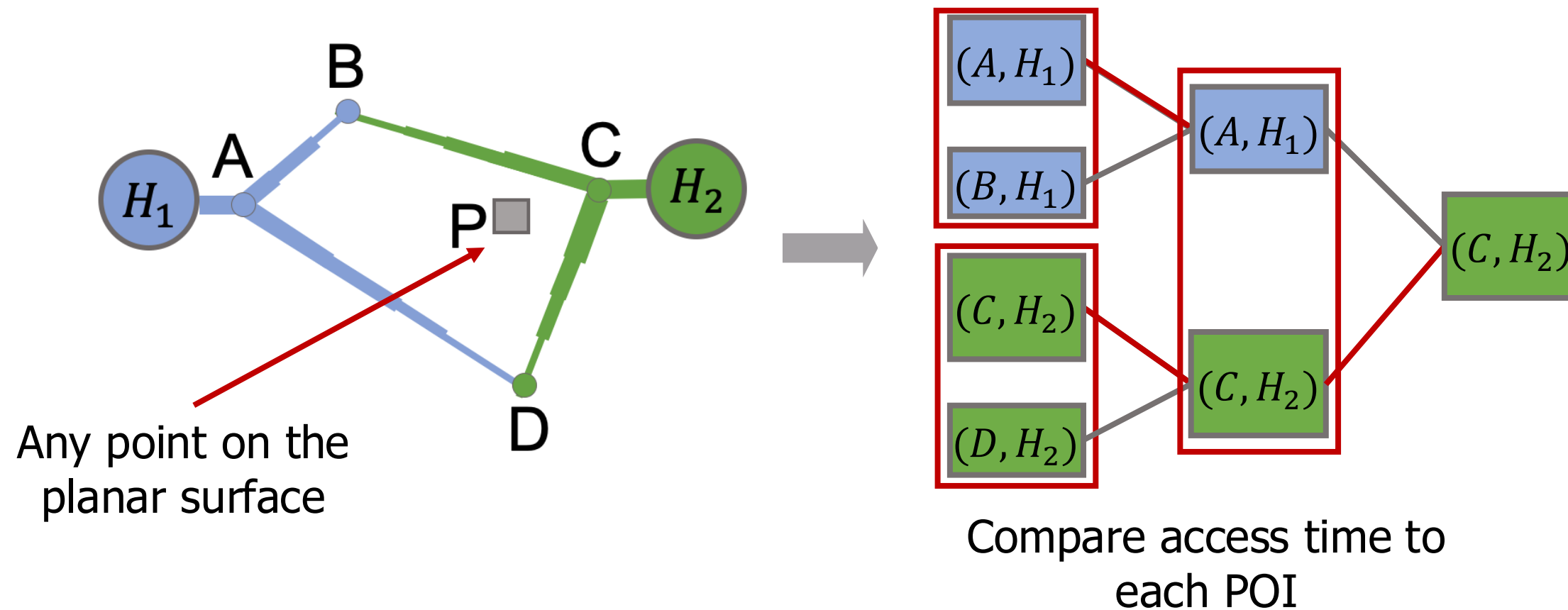


Find the shortest path

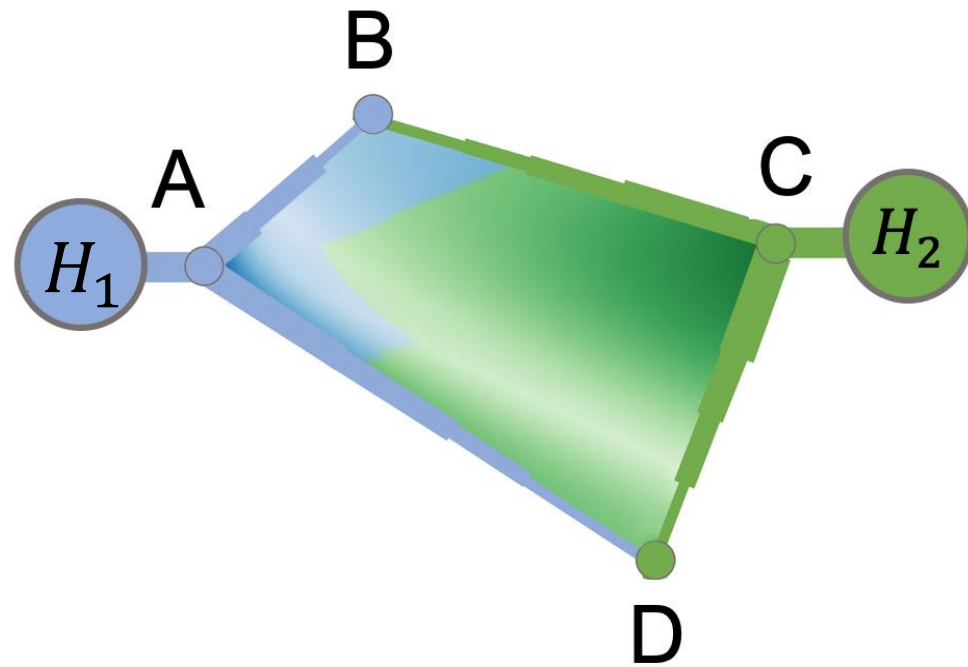


Compute density field on each vertex and draw the density along the edge

Density Estimation on 2D Planar Surface



Density Estimation on 2D Planar Surface Cont'd

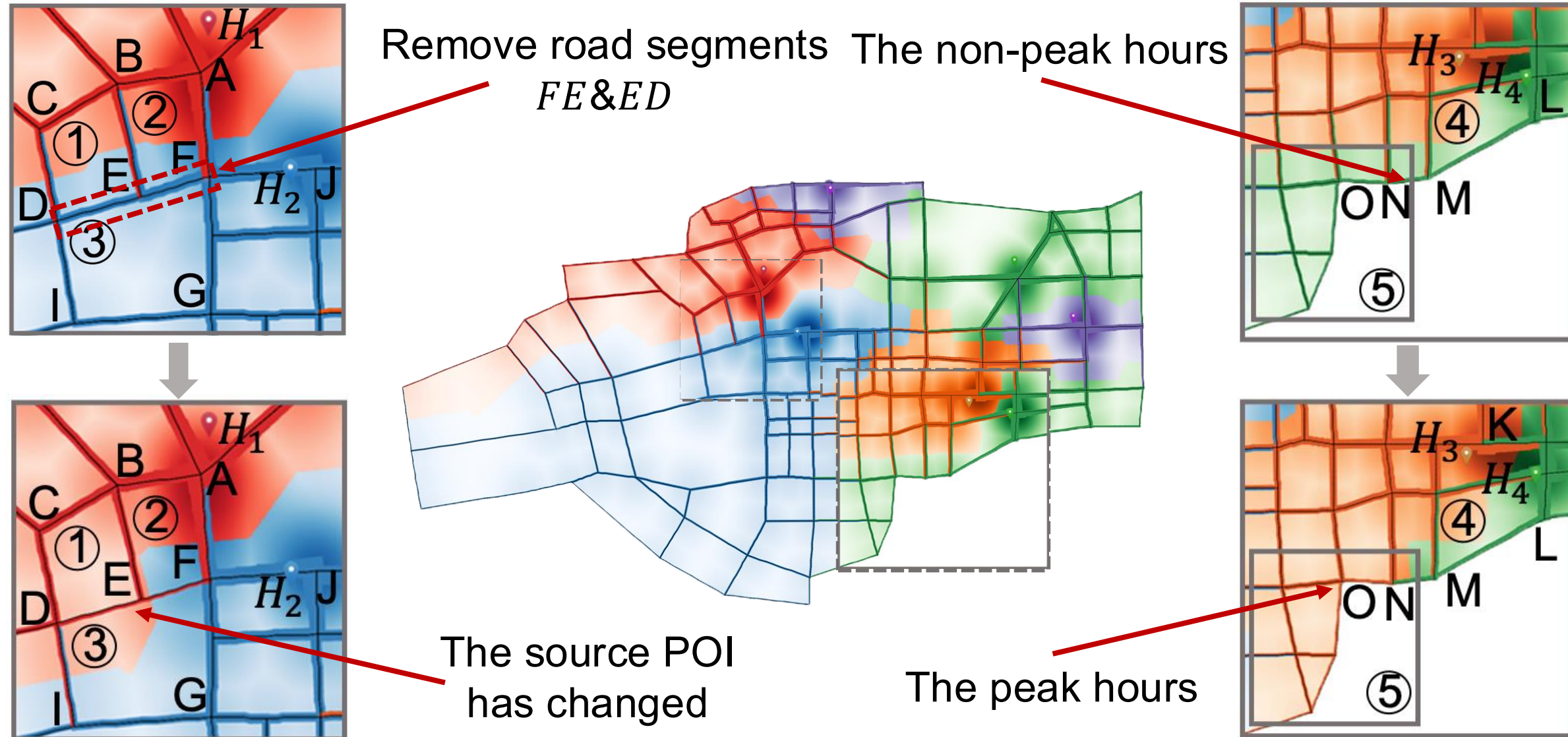


Estimate density field

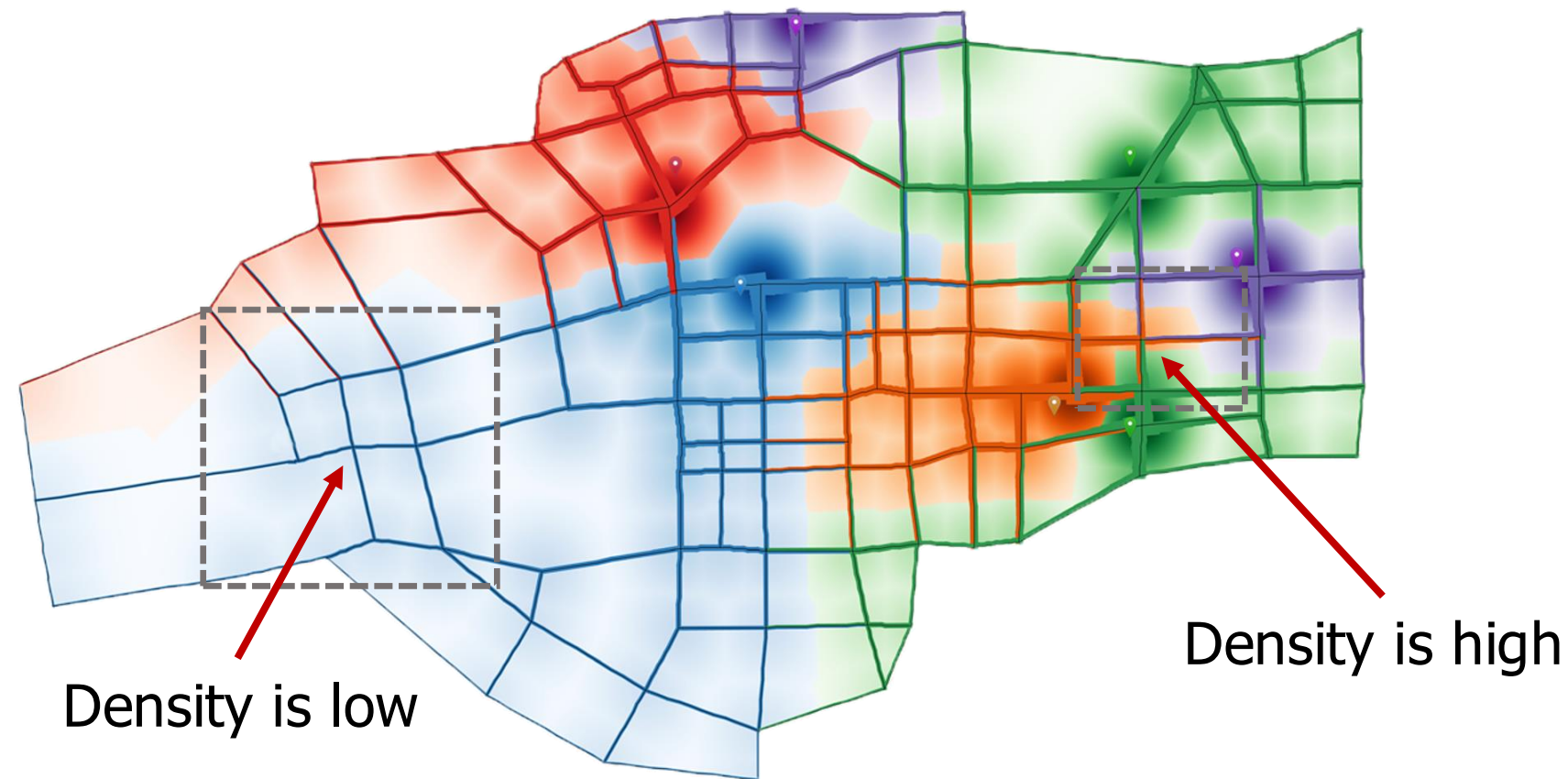
$$\lambda(P|H_2, G) = \frac{1}{r} K\left(\frac{(t(H_2C) + t(CP))^{-\alpha}}{r}\right)$$

$\lambda(P|H_2, G)$: Density field at P given POI H_2 and road network G
 $(t(H_2C) + t(CP))^{-\alpha}$: H_2 accessibility at location P
 K : Kernel function
 r : kernel radius

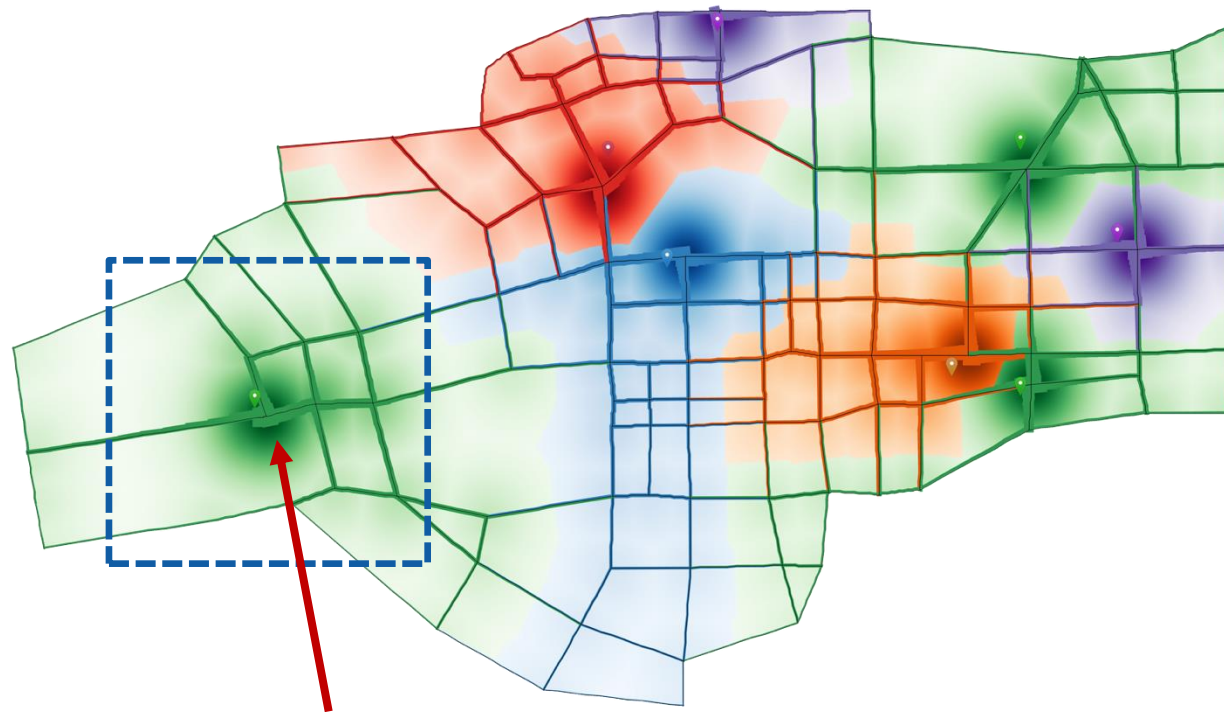
Study 1: The Influence of Road Topology and Traffic Conditions on Accessibility



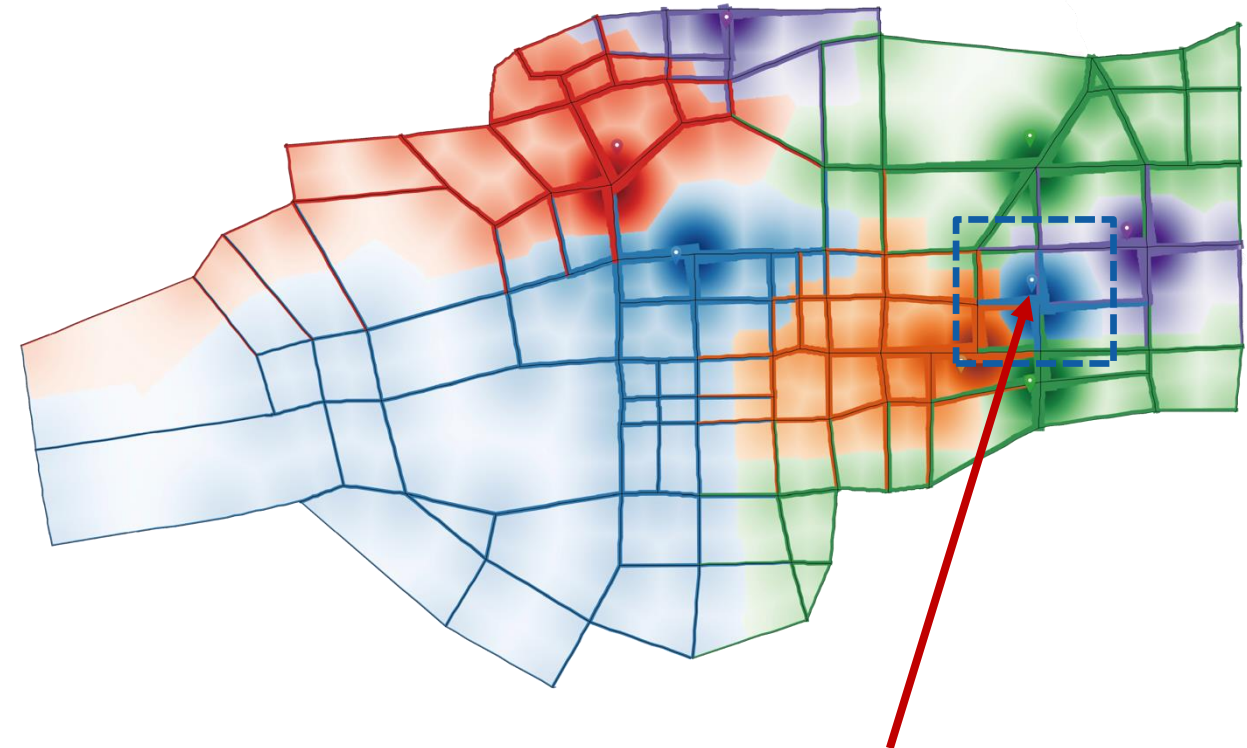
Study 2: Finding the Optimal Location of a New Public Service Facility



Study 2: Finding the Optimal Location of a New Public Service Facility



After adding new POI



After adding new POI

Computation Efficiency

Table.1 Running time (in milliseconds) comparison for topology density map and planar KDE by different number of POIs and image

Image Size	10 POIs		50 POIs		100 POIs	
	Topology	Planar	Topology	Planar	Topology	Planar
1920×1200	2010.75	99.41	1992.04	144.21	2134.92	192.20
1280×800	962.55	51.25	971.14	78.49	991.41	116.29
640×400	353.87	18.98	381.97	24.33	347.03	32.70

- The computation efficiency for planar KDE is affected by both image size and number of POIs
- The computation efficiency for topology density map is affected by image size
- The computation efficiency for topology density map is also affected by the number of intersections rather than the number of POIs

Limitation & Future Work

Limitation:

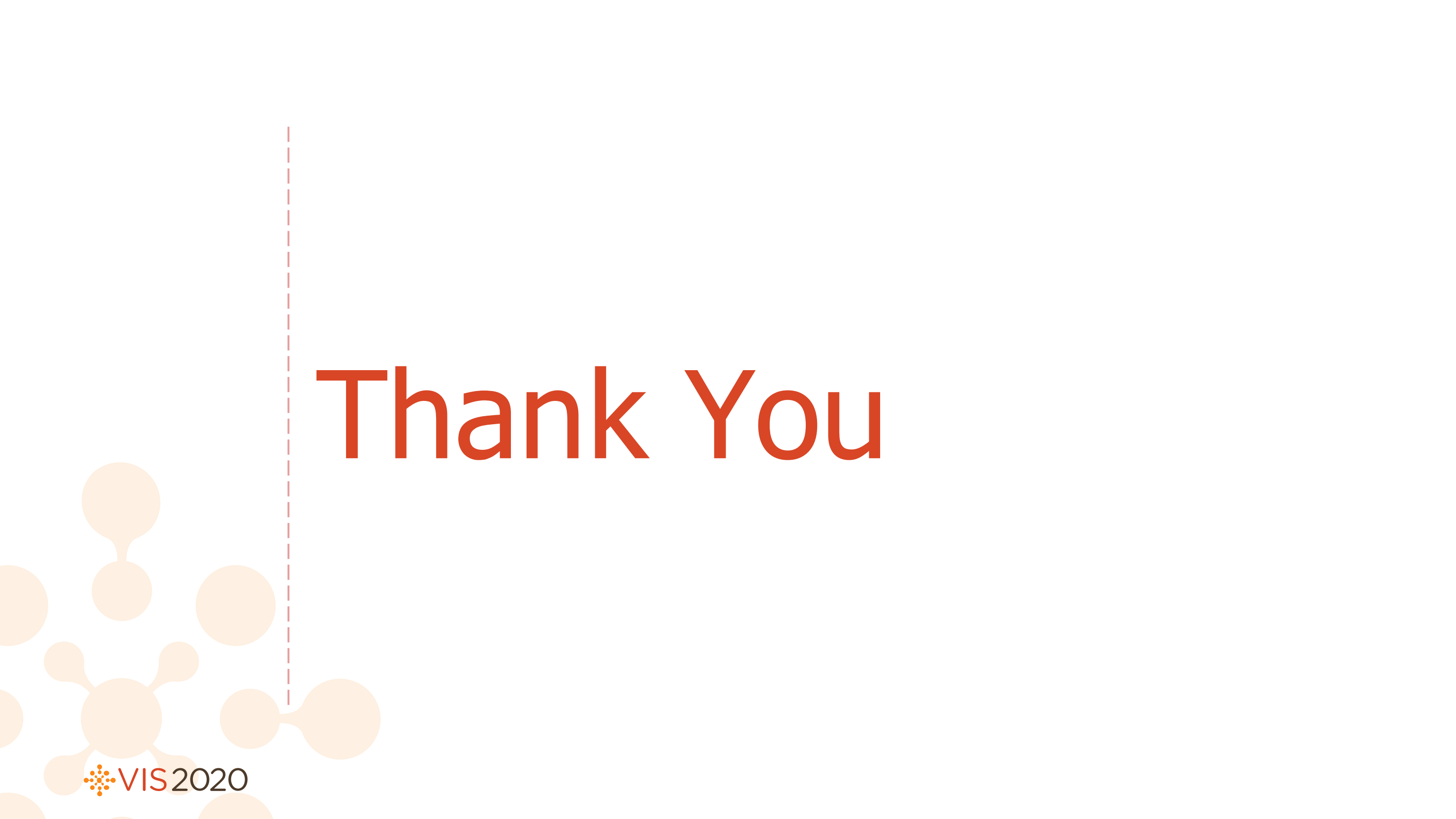
- The computational efficiency of generating the density map is not high
- More fine-grained data is needed
- Lacking the building and walkway data in each zone

Future work:

- Integrate topology density map into many real-world applications
- Provide temporal information in the context of a map view
- Combine with other dataset such as climate and population data to do more exploration

Take Home Messages

- ✓ The topology of the road network in an urban area is one of a critical factor in urban analysis
- ✓ A novel approach to generate density map taking road topology into consideration
- ✓ After combining different dataset, our method can be used to analyze other urban problems related to movement pattern



Thank You